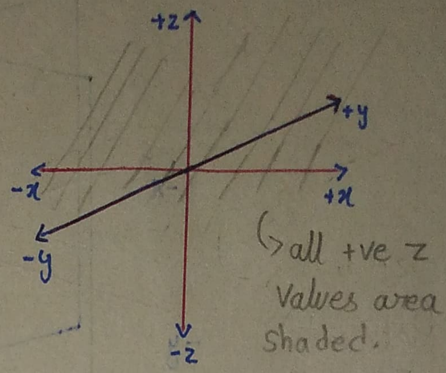
Multi-Variable Calculus
Assignment 1

Q1i) $f(x,y,z) = xy\ln z$

$z > 0$

$\therefore$ Domain $= \{(x,y,z) \in \mathbb{R}^3 \mid z > 0\}$

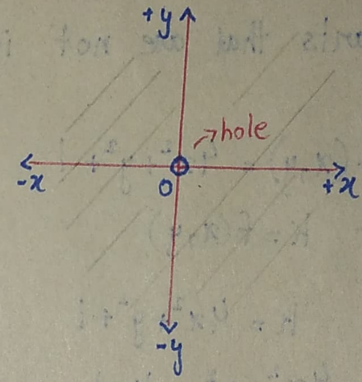


Note:-
(The graph sketched will be above $z > 0$ and x, y can be any values. No negative z value)

Q1ii) $f(x,y) = \ln(x^2+y^2)$

$x^2+y^2 > 0$

$\therefore$ Domain $= \{(x,y) \in \mathbb{R}^2 \mid (x,y) \neq (0,0)\}$

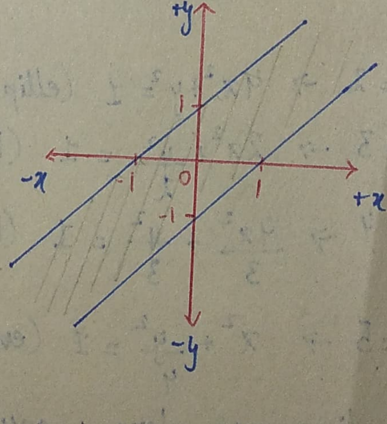


Note:-
(Whole domain is shaded except point $(0,0)$ where there is a hole).

Q1iii) $f(x,y) = \sin^{-1}(y-x)$

$-1 \leq y-x \leq 1$

$\therefore$ Domain $= \{(x,y) \in \mathbb{R}^2 \mid x-1 \leq y \leq x+1$



Q1iv) $f(x,y,z) = \dfrac{1}{x+1} + \dfrac{1}{y-1} + \dfrac{1}{x+y-z}$

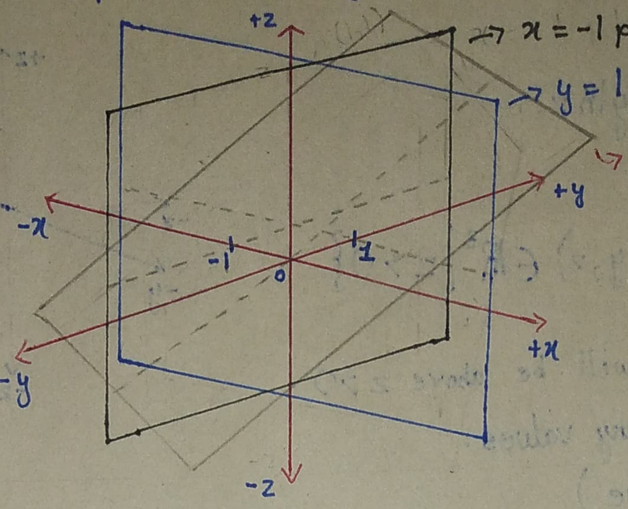
$x+1 \neq 0$	$y-1 \neq 0$	$x+y-z \neq 0$
$x \neq -1$	$y \neq 1$	$z \neq x+y$

$\therefore$ Domain $\{(x,y,z) \in \mathbb{R}^3 \mid x \neq -1, y \neq 1, z \neq x+y\}$

The domain is the entire 3D space except these 3 planes: $x = -1$ (a vertical plane parallel to the yz-plane)

$y=1$ (a vertical plane parallel to the xz-plane)

$z=x+y$ (a tilted plane intersecting all 3 axis)



The dotted pencil line Show's the plane passing through proper points. (Makes the graph more easy to read in my opinion)

The parts that are not in plane is part of domain.

Q2i) $f(x,y) = 4x^2+y^2+1$

let $k = f(x,y)$

so

$$k = 4x^2+y^2+1$$

$$4x^2+y^2 = k-1$$

$$\frac{4x^2}{(k-1)} + \frac{y^2}{(k-1)} = 1$$

when

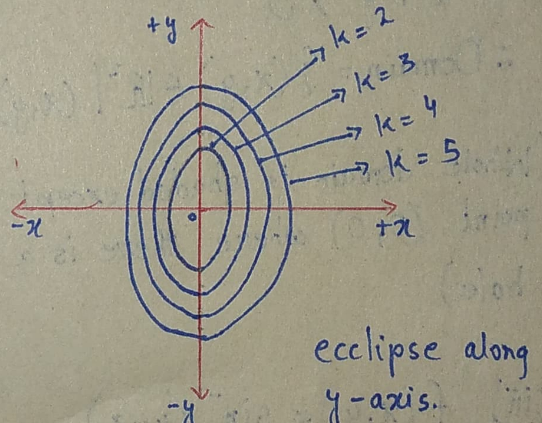
$k=2 \rightarrow 4x^2+y^2 = 1$ (ellipse)

$k=3 \rightarrow 2x^2+\frac{y^2}{2} = 1$ (large ellipse)

$k=4 \rightarrow \frac{4x^2}{3} + \frac{y^2}{3} = 1$ (larger eclipse)

$k=5 \rightarrow x^2+\frac{y^2}{4} = 1$ (even larger eclipse)

The ecclipse are along y-axis



Q2ii) $-2 = 2x - 6y + z$

$z = 6y - 2x - 2$

let $z = k$

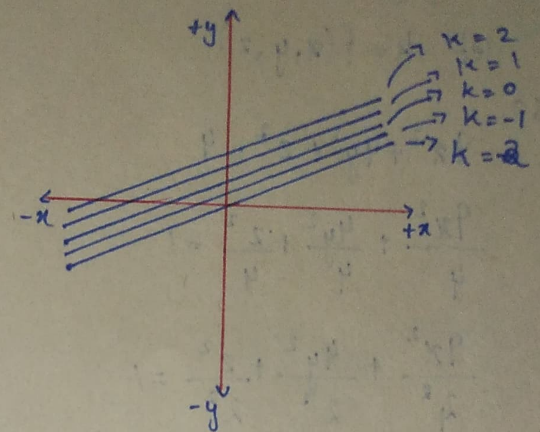
so

$k = 6y - 2x - 2$

$k + 2 = 6y - 2x$

$y = \frac{2x}{6} + \frac{k+2}{6}$

$y = \frac{x}{3} + \frac{k+2}{6}$



This is an eq of straight line with slope $\frac{x}{3}$ and y-intercept at $\frac{k+2}{6}$

when

$k = 0 \rightarrow y = \frac{1}{3}x + \frac{1}{3}$

$k = 1 \rightarrow y = \frac{1}{3}x + \frac{1}{2}$

$k = 2 \rightarrow y = \frac{1}{3}x + \frac{2}{3}$

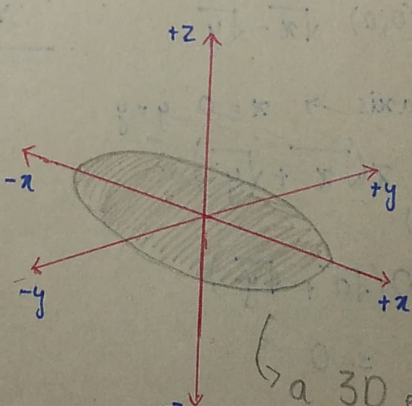
$k = -2 \rightarrow y = \frac{1}{3}x + 0$

$k = -1 \rightarrow y = \frac{1}{3}x + \frac{1}{6}$

Q3i) $\frac{x^2}{25} + \frac{y^2}{16} + \frac{z^2}{9} = k$

when k is sent to L.H.S and $k > 0$

$\frac{x^2}{25k} + \frac{y^2}{16k} + \frac{z^2}{9k} = 1$



a 3D ellopsoid sketch.

- as denominator value are not same it is an ellopsoid

$\frac{x^2}{\sqrt{k}5^2} + \frac{y^2}{\sqrt{k}4^2} + \frac{z^2}{\sqrt{k}3^2} = 1$

$\frac{x^2}{5\sqrt{k}} + \frac{y^2}{4\sqrt{k}} + \frac{z^2}{3\sqrt{k}} = 1$

- if $k = 0$ than equation reduces to $x = y = z = 0$, which is origin
- if $k < 0$, there are no real solution so the level surface doesn't exis.

for $k > 0$

- the ellopsoid is stretched most along x-axis.
- less on y-axis
- and least on z-axis

Q3ii) $f(x,y,z) = 9x^2 + 4y^2 + z^2 \; ; \; k = 4$

let $k = f(x,y,z)$

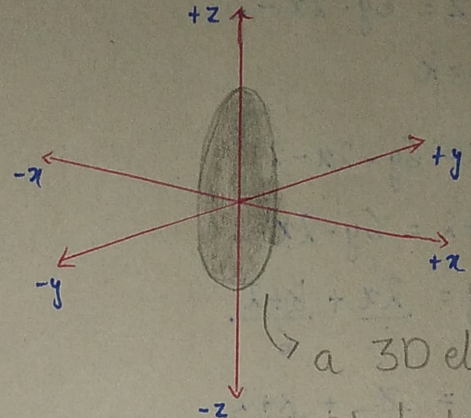
so

$$9x^2 + 4y^2 + z^2 = 4$$

$$\frac{9x^2}{4} + \frac{4y^2}{4} + \frac{z^2}{4} = 1$$

$$\frac{9x^2}{4} + \frac{4y^2}{2^2} + \frac{z^2}{2^2} = 1$$

$$\frac{x^2}{\left(\frac{2}{3}\right)^2} + \frac{y^2}{1^2} + \frac{z^2}{2^2} = 1$$



a 3D ellopsoid Sketch

as denominator isn't same it is an ellopsoid.

- where ellopsoid is stretched more along z-axis (value = 2)
- then less along y-axis (value = 1)
- then least along x-axis (value = $\frac{2}{3}$)

Q4i) $\lim\limits_{(x,y)\to(0,0)} \dfrac{x^2 - xy}{\sqrt{x} - \sqrt{y}}$ → simplified version

$$\frac{x(x-y)\cdot(\sqrt{x}+\sqrt{y})}{(\sqrt{x}-\sqrt{y})(\sqrt{x}+\sqrt{y})}$$

$$\frac{x\,\cancel{(x-y)}(\sqrt{x}+\sqrt{y})}{\cancel{(x-y)}}$$

$$x(\sqrt{x}+\sqrt{y})$$

$\therefore \lim\limits_{(x,y)\to(0,0)} 0(\sqrt{0}+\sqrt{0})$

$= 0$ Ans

along y-axis → $x = 0$ $y = y$

~~$\lim\limits_{(0,y)\to(0,0)} x(\sqrt{x}+\sqrt{y})$~~

~~$\lim\limits_{y\to 0} 0(\sqrt{0}+\sqrt{y})$~~

~~$= 0$~~

~~along $y = 0$~~

~~$x(\sqrt{x}+\sqrt{y})$~~

~~$x(\sqrt{x}+\sqrt{0})$~~

confirmation check

along $y = x$

$\lim\limits_{(x,y)\to(0,0)} x(\sqrt{x}+\sqrt{y})$

$\lim\limits_{(x,x)\to(0,0)} x(\sqrt{x}+\sqrt{x})$

$\lim\limits_{(x,x)\to(0,0)} x(2\sqrt{x})$

apply limit

$0(2\sqrt{0})$

$= 0$ hence proved limit exist

Q4ii) $\lim_{(x,y)\to(0,0)} \cos\left(\frac{x^2-y^2}{x+y+1}\right)$

$= \cos\left(\frac{0^2-0^2}{0+0+1}\right)$

$= \cos\left(\frac{0}{1}\right)$

$= \cos(0)$

$= 1$

let's confirm along $y = x^2$

$f(x, x^2) \Rightarrow \cos\left(\frac{x^2-x^5}{x+x^2+1}\right)$

$= \lim_{x\to 0} \cos\left(\frac{x^2-x^5}{x+x^2+1}\right)$

$= \cos\left(\frac{0^2-0^5}{0+0^2+1}\right)$

$= \cos\left(\frac{0}{1}\right)$

$= \cos(0)$

$= 1$

hence limit exist

Q5) $V = \frac{1}{3}\pi r^2 h$

$h = 3.2cm$ or $32mm$

$r = 1.5cm$ or $15mm$

$\frac{dh}{dt} = 3mm/s$, $\frac{dr}{dt} = -2m/s$, $\frac{dV}{dt} = ??$

$\frac{dV}{dt} = \frac{1}{3}\pi\left(2rh.\frac{dr}{dt} + r^2\frac{dh}{dt}\right)$

$\frac{dV}{dt} = \frac{1}{3}\pi\left(2(15)(32)(-2) + (15)^2(3)\right)$

$\frac{dV}{dt} = -1303.7$

$\frac{dV}{dt} \simeq -1300\ mm^3/s$

Q6) $A = \frac{1}{2} ac\sin B$

$a = 3$ unit

$B = \frac{\pi}{6}$ radian

$c = 4$ unit

$\frac{da}{dt} = 0.4 unit/s$, $\frac{dc}{dt} = -0.8 unit/s$, $\frac{dB}{dt} = 0.2 rad/s$

$\frac{dA}{dt} = \frac{1}{2}\left(c\sin B.\frac{dA}{dt} + a\sin B.\frac{dc}{dt} + ac\cos B\frac{dB}{dt}\right)$

$= \frac{1}{2}\left(4\sin\frac{\pi}{6}(0.4) + 3\sin\frac{\pi}{6}(-0.8) + 3(4)\cos\frac{\pi}{6}\cdot(0.2)\right)$

$= 0.839\ units^2/s$

Q7) $pV^{1.4} = k$

$\ln(p) + 1.4\ln(V) = \ln(k)$

Differentiate

$\frac{1}{p} \cdot dp + \frac{1.4\, dV}{V} = \frac{1}{k} dk$

$\frac{dk}{k} = \frac{dp}{p} + (1.4)\frac{dV}{V}$

$\frac{dp}{p} = 4\% = 0.04$

$\frac{dV}{V} = -1.5\% = -0.015$

$\frac{dk}{k} = 0.04 + 1.4(-0.015)$

$\frac{dk}{k} = 0.019$

$\therefore$ approximate percentage error in k is $0.019 \times (100) = 1.9\%$

Q8) $W = f(T, v)$

$f_T(-15, 30)$

$\Delta T = 5$

$f_T = \frac{W(T_0 + \Delta T, v_0) - W(T_0, v_0)}{\Delta T}$

$= \frac{W(-15+5, 30) - W(-15, 30)}{5}$

$= \frac{W(-10, 30) - W(-15, 30)}{5}$

$= \frac{-20 - (-26)}{5}$

$= \frac{6}{5}$

$= 1.2$

when $\Delta T = -5$

$f_T = \frac{W(T_0 + \Delta T, v_0) - W(T_0, v_0)}{\Delta T}$

$= \frac{W(-15-5, 30) - W(-15, 30)}{-5}$

$= \frac{W(-20, 30) - W(-15, 30)}{-5}$

$= \frac{-33 - (-26)}{-5}$

$= \frac{7}{5}$

$= 1.4$

$\text{avg} = \frac{1.2 + 1.4}{2}$

$= 1.3$

$W = f(T, v)$

$f_v(-15, 30)$

$\Delta v = 10$

$$f_v = \frac{W(T_0, v_0 + \Delta v) - W(T_0, v_0)}{\Delta v}$$

$$= \frac{W(-15, 30+10) - W(-15, 30)}{10}$$

$$= \frac{W(-15, 40) - W(-15, 30)}{10}$$

$$= \frac{-27 - (-26)}{10}$$

$$= -0.1$$

when

$\Delta v = -10$

$$f_v = \frac{W(T_0, v_0 + \Delta v) - W(T_0, v_0)}{\Delta v}$$

$$= \frac{W(-15, 30-10) - W(-15, 30)}{-10}$$

$$= \frac{W(-15, 20) - W(-15, 30)}{-10}$$

$$= \frac{-24 - (-26)}{-10}$$

$$= -0.2$$

$$avg = \frac{-0.1 + (-0.2)}{2}$$

$$= -0.15$$

Practical Applications

1-) $f_T(-15, 30) = 1.3$ means that for a fixed wind speed of 30 km/h, the percieved temperature increase by 1.3°C for each 1° increase in actual temperature.

2-) for each 1 km/h increase in wind speed at $T = -15°C$, the percieved temperature decreases by 0.15°C.

Q9) $h = f(v, t)$

a) $\frac{\partial h}{\partial v}$ mean partial derivative of 'h' with respect to 'v' and $\frac{\partial h}{\partial t}$ is partial derivative of 'h' with respect to 't'.

9b) $f_v(40, 15)$

$\Delta v = 10$

$$f_v = \frac{h(v_0 + \Delta v, t_0) - h(v_0, t_0)}{\Delta v}$$

$$= \frac{h(40+10, 15) - h(40, 15)}{10}$$

$$= \frac{h(50, 15) - h(40, 15)}{10}$$

$$= \frac{36 - 25}{10}$$

$$= 1.1$$

$\Delta v = -10$

$$f_v = \frac{h(v_0 + \Delta v, t_0) - h(v_0, t_0)}{\Delta v}$$

$$= \frac{h(40-10, 15) - h(40, 15)}{-10}$$

$$= \frac{h(30, 15) - h(40, 15)}{-10}$$

$$= \frac{16 - 25}{-10}$$

$$= 0.9$$

$$\text{avg} = \frac{1.1 + 0.9}{2}$$

$$= 1$$

$f_T(40, 15)$

$\Delta t = 5$

$$f_T = \frac{h(v_0, t_0 + \Delta t) - h(v_0, t_0)}{\Delta t}$$

$$= \frac{h(40, 15+5) - h(40, 15)}{5}$$

$$= \frac{h(40, 20) - h(40, 15)}{5}$$

$$= \frac{28 - 25}{5}$$

$$= 0.6$$

when $\Delta t = -5$

$$f_T = \frac{h(v_0, t_0 + \Delta t) - h(v_0, t_0)}{\Delta t}$$

$$= \frac{h(40, 15-5) - h(40, 15)}{-5}$$

$$= \frac{h(40, 10) - h(40, 15)}{-5}$$

$$= \frac{21 - 25}{-5}$$

$$= 0.8$$

$$\text{avg} = \frac{0.6 + 0.8}{2}$$

$$= 0.7$$

Practical Application

1) $f_v(40, 15)$: for every 1 knot increase in wind speed at $t = 15$ hours, the wave height increases by 1 feet.

2) $f_t(40, 15)$: for every 5-hour increase in wind blowing time at $v = 40$ knots, the wave height increases by 0.7 feet.

Q10 i) $f(x,y,z) = xe^y + ye^z + ze^x$, $(0,0,0)$, $v = \{5,1,-2\}$

$$\frac{\hat{v}}{|v|} = \frac{5i+j-2k}{\sqrt{5^2+1^2+(-2)^2}} = \frac{5i}{\sqrt{30}} + \frac{j}{\sqrt{30}} - \frac{2k}{\sqrt{30}}$$

$$D_u f(x,y,z) = f_x \cdot u_1 + f_y \cdot u_2 + f_z \cdot u_3$$

$f_x = e^y + 0 + ze^x = e^y + ze^x$ | $f_x(0,0,0) = e^0 + 0e^0 = 1$

$f_y = xe^y + e^z + 0 = xe^y + e^z$ | $f_y(0,0,0) = 0e^0 + e^0 = 1$

$f_z = 0 + ye^z + e^x = ye^z + e^x$ | $f_z(0,0,0) = 0e^0 + e^0 = 1$

$$\therefore D_u f(x,y,z) = 1\left(\frac{5}{\sqrt{30}}\right) + 1\left(\frac{1}{\sqrt{30}}\right) - 1\left(\frac{2}{\sqrt{30}}\right)$$

$$= \frac{4}{\sqrt{30}} \quad \text{Ans}$$

ii) $f(x,y,z) = \sqrt{xyz}$, $(3,2,6)$, $v = \{-1,-2,2)$

$$\frac{\hat{v}}{|v|} = \frac{-i-2j+2k}{\sqrt{(-1)^2+(-2)^2+2^2}} = \frac{-i}{3} - \frac{2j}{3} + \frac{2k}{3}$$

$$D_u f(x,y,z) = f_x \cdot u_1 + f_y \cdot u_2 + f_z \cdot u_3$$

$f_x = \frac{yz}{2\sqrt{xyz}}$ $\qquad f_x(3,2,6) = \frac{2(6)}{2\sqrt{3(2)(6)}} = 1$

$f_y = \frac{xz}{2\sqrt{xyz}}$ $\qquad f_y(3,2,6) = \frac{3(6)}{2\sqrt{3(2)(6)}} = \frac{3}{2}$

$f_z = \frac{xy}{2\sqrt{xyz}}$ $\qquad f_z(3,2,6) = \frac{3(2)}{2\sqrt{3(2)(6)}} = \frac{1}{2}$

$$\therefore D_u f = 1\left(-\frac{1}{3}\right) + \frac{3}{2}\left(\frac{-2}{3}\right) + \frac{1}{2}\left(\frac{2}{3}\right)$$

$$= -1 \quad \text{Ans}$$

Q10 iii) $f(x,y) = x - \frac{y^2}{x} + \sqrt{3}\sec^{-1}(2xy)$; $(1,1)$, $v = \{12i + 5j\}$

$$\frac{\hat{v}}{|v|} = \frac{12i+5j}{\sqrt{12^2+5^2}} = \frac{12i}{13} + \frac{5j}{13}$$

$\boxed{\text{derivative of } \sec^{-1}(x) \text{ is } \frac{1}{|x|\sqrt{x^2-1}}}$

$$f_x = 1 + \frac{y^2}{x^2} + \sqrt{3}\left[\frac{2y}{|2xy|\sqrt{(2xy)^2-1}}\right]$$

$$= 1 + \frac{y^2}{x^2} + \frac{\sqrt{3}(2y)}{|2xy|\sqrt{4x^2y^2-1}}$$

put $f_x(1,1)$

$$= 1 + \frac{1^2}{1^2} + \frac{\sqrt{3}(2(1))}{|2(1)(1)|\sqrt{4(1)^2(1)^2-1}}$$

$$= 1 + 1 + \frac{2\sqrt{3}}{|2|\sqrt{3}}$$

$$= 1 + 1 + 1$$

$$= 3$$

$$f_y = \frac{-2y}{x} + \frac{\sqrt{3}\cdot 2x}{|2xy|\sqrt{4x^2y^2-1}}$$

put $f_y(1,1)$

$$= \frac{-2(1)}{(1)} + \frac{\sqrt{3}\cdot 2(1)}{|2(1)(1)|\sqrt{4(1)^2(1)^2-1}}$$

$$= -2 + \frac{2\sqrt{3}}{|2|\sqrt{3}}$$

$$= -1$$

$\therefore D_u f = f_x \cdot U_1 + f_y \cdot U_2$

$$= 3\left(\frac{12}{13}\right) + (-1)\left(\frac{5}{13}\right)$$

$$= \frac{31}{13} \quad \text{Ans}$$